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Winding Device

Abstract

The disclosed subject matter provides a winding device that includes a material-placing mechanism, a mounting table, a winding member and a restraining member, wherein the material-placing mechanism is configured to convey a material strip; the winding member includes a winding needle base and a plurality of winding needles provided on the winding needle base, the winding needle base is rotatably provided on the mounting table, with a winding needle gap formed between the plurality of winding needles, such that the material strip passes through; the restraining member is provided outside the winding needle gap and configured to fix the material strip protruding from the winding needle gap with a preset force F1, the winding needle base is configured to drive the winding needle to rotate with a winding tension F2, and F1 is less than F2.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] The present disclosure is a continuation of International Application No. PCT/CN2023/131327, filed on Nov. 13, 2023, which claims priority to a Chinese patent application No. 202223106651.8 filed with the CNIPA on Nov. 22, 2022 and entitled “WINDING DEVICE”, both of which are hereby incorporated by reference in their entireties.

TECHNICAL FIELD

[0002] The present disclosure relates to the technical field of cell processing equipment, and particularly to a winding device.

BACKGROUND

[0003] With the development of battery technology, there is an increasing demand for battery miniaturization. As the core component of a battery, a cell must also be miniaturized. Consequently, the demand for high processing quality for miniaturized cells has also been continuously rising.

[0004] In the relevant technology, a clamping member, such as an inner clamping needle, is provided on a winding needle. This inner clamping needle can clamp and fix the initial winding end of the material strip onto the winding needle before the material strip is wound onto the winding needle, thereby preventing the material strip from sliding relative to the winding needle. In the case that the material strip can be prevented from sliding relative to the winding needle by the inner clamping needle, the material strip can be wound neatly onto the winding needle when the winding needle base starts to drive the winding needle to rotate so as to wind the material strip onto the winding needle.

[0005] However, when winding a small-sized cell, the winding needle that is adapted to the small cell must also be small. Due to size constraints, it is not feasible to provide a clamping member on the winding needle to fix the initial winding end of the material strip. Consequently, when the winding needle base starts to drive the winding needle to rotate, the material strip may slip relative to the winding needle. This causes the subsequent winding to be disordered and uneven, resulting in suboptimal winding performance.

SUMMARY

[0006] In view of the above deficiencies in the prior art, the present disclosure provides a winding device, which can ensure the winding performance for small-sized cells.

[0007] In order to solve the above technical problem, the present disclosure provides a winding device, which includes: [0008] a material-placing mechanism configured to convey a material strip; [0009] a mounting table; [0010] a winding member including a winding needle base and a plurality of winding needles provided on the winding needle base, the winding needle base being rotatably provided on the mounting table, with a winding needle gap formed between the plurality of winding needles and configured to allow the material strip to pass through; and [0011] a restraining member provided outside the winding needle gap and configured to fix the material strip protruding from the winding needle gap with a preset force F_1 , where a winding tension applied to the material strip when the winding needle base drives the winding needle to rotate is F_2 , and F_1 is less than F_2 , so as to wind both a preset length of the material strip and the material strip protruding

from the winding needle gap onto the winding needle when the winding needle base drives the winding needle to rotate.

[0012] Optionally, the restraining member is movable towards or away from the winding needle so as to fix the material strip protruding from the winding needle gap with the preset force or release it.

[0013] Optionally, the restraining member includes a first driving member and an adsorption member connected to the first driving member, the first driving member is configured to drive the adsorption member to move towards or away from the winding needle, and the adsorption member is configured to adsorb and fix the material strip protruding from the winding needle gap.

[0014] Optionally, the adsorption member is an adsorption plate, which is provided with a plurality of first adsorption holes on a side of the adsorption plate facing the material strip.

[0015] Optionally, the restraining member includes a second driving member and a pressure roller rotatably connected to the second driving member, the second driving member is configured to drive the pressure roller to move towards or away from the winding needle, and the pressure roller is configured to press the material strip protruding from the winding needle gap onto the winding needle so as to fix the material strip by pressing.

[0016] Optionally, the restraining member further includes a mounting base connected to the second driving member, and an elastic member provided on the mounting base, with the pressure roller provided on the elastic member.

[0017] Optionally, the mounting table includes a rotating plate, which is rotatable about its central axis; [0018] the winding member includes a first winding member and a second winding member; [0019] the winding device has a winding station and a material-unloading station, when the first winding member winds a preset length of the material strip at the winding station, the rotating plate is configured to drive the first winding member to rotate to the material-unloading station and to drive the second winding member to rotate to the winding station, and the winding needle base of the second winding member is further configured to drive the winding needle to slide and protrude out relative to the rotating plate along the axial direction of the winding needle, so as to enable the material strip to be located in the winding needle gap of the second winding member; [0020] the winding device further includes a cutter assembly, which is configured to cut off a material strip between the first winding member and the second winding member, and the restraining member is configured to fix the material strip which protrudes from the winding needle gap and has been cut off.

[0021] Optionally, the cutter assembly includes a cutter driving member and a cutter connected to the cutter driving member, and the cutter driving member is configured to drive the cutter to move towards the material strip so as to cut off the material strip between the first winding member and the second winding member.

[0022] Optionally, the cutter is provided with a heating rod therein, which is configured to heat the cutter.

[0023] Optionally, the mounting table further includes a connection shaft and an abutting member provided on the connection shaft, the abutting member is configured to provide an abutting force to the material strip when the cutter cuts off the material strip between the first winding member and the second winding member, and the rotating plate is fixed to the connection shaft.

[0024] Optionally, the abutting member is provided with an avoidance groove, which is configured to avoid the cutter.

[0025] Optionally, the abutting member, on a side facing the material strip, is provided with a first adsorption plate and a second adsorption plate, there is an avoidance gap between the first adsorption plate and the second adsorption plate, the avoidance gap forms the avoidance groove, and the first adsorption plate and the second adsorption plate are configured to adsorb the material strip.

[0026] Optionally, the winding device further includes a finishing assembly, and the finishing

assembly includes a finish compressing member provided adjacent to the material-unloading station, and is configured to press the material strip, which is adjacent to the material-unloading station and has been cut off, onto the winding member at the material-unloading station.

[0027] Optionally, the finishing assembly further includes a finishing adsorption member provided adjacent to the material-unloading station, and the finishing adsorption member is configured to adsorb the material strip, which is adjacent to the material-unloading station and has been cut off, so as to prevent a free end of the material strip from swinging.

[0028] Compared with the prior art, the present disclosure at least achieves the following beneficial effects:

[0029] In the present disclosure, since the restraining member fixes the material strip protruding from the winding needle gap with a preset force, and the preset force is less than the winding tension applied to the material strip when the winding needle base drives the winding needle to rotate, when the winding needle base drives the winding needle to rotate relative to the mounting table, it is possible to prevent the material strip protruding from the winding needle gap from sliding relative to the winding needle, and at the same time to ensure that the material strip protruding from the winding needle gap is wound onto the winding needle, thereby ensuring the neatness of the material strip wound onto the winding needle and improving the winding performance of the material strip. Additionally, since the restraining member is provided outside the winding needle gap, the winding device can be adapted for winding a small-sized cell.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0030] To better illustrate technical solutions of embodiments of the present disclosure, the accompanying drawings required for the embodiments will be briefly introduced as follows. Note that the drawings described here are only some embodiments of the present disclosure. For those skilled in the art, additional drawings can be derived based on these disclosed drawings without creative efforts.

[0031] FIG. 1 is a structural schematic diagram of a winding device provided by an embodiment of the present disclosure;

[0032] FIG. 2 is a structural schematic diagram of a winding member provided by an embodiment of the present disclosure;

[0033] FIG. 3 is a structural schematic diagram of a restraining member being an adsorption member provided by an embodiment of the present disclosure;

[0034] FIG. 4 is a structural schematic diagram of an adsorption plate provided by an embodiment of the present disclosure;

[0035] FIG. 5 is a structural schematic diagram when a restraining member provided by an embodiment of the present disclosure is a pressure roller and a material strip is not pressed by the pressure roller;

[0036] FIG. 6 is a structural schematic diagram when a restraining member provided by an embodiment of the present disclosure is a pressure roller and a material strip is pressed by the pressure roller;

[0037] FIG. 7 is a structural schematic diagram of a connection of a pressure roller provided by an embodiment of the present disclosure;

[0038] FIG. 8 is a structural schematic diagram of configurations of a first winding member and a second winding member provided by an embodiment of the present disclosure;

[0039] FIG. 9 is a structural schematic diagram of a cutter assembly provided by an embodiment of the present disclosure;

[0040] FIG. 10 is a structural schematic diagram of a configuration of an abutting member

provided by an embodiment of the present disclosure;

[0041] FIG. **11** is a front view of the abutting member provided by an embodiment of the present disclosure;

[0042] FIG. **12** is a structural schematic diagram of a finishing assembly provided on the winding device provided by an embodiment of the present disclosure;

[0043] FIG. **13** is a schematic diagram of the finishing assembly provided by an embodiment of the present disclosure.

DESCRIPTION OF REFERENCE SIGNS

[0044] **10**—material strip; [0045] **100**—winding device; **110**—material-placing mechanism; [0046] **120**—mounting table; **121**—rotating plate; **122**—abutting member; **1221**—avoidance groove; **1222**—first adsorption plate; **1223**—second adsorption plate; **1224**—second adsorption hole; **123**—connection shaft; [0047] **130**—winding member; **131**—winding needle base; **132**—winding needle; **133**—winding needle gap; **134**—first winding member; **135**—second winding member; [0048] **140**—restraining member; **141**—adsorption member; **1411**—first driving member; **1412**—first adsorption hole; **142**—pressure roller; **1421**—second driving member; **1422**—mounting base; **1423**—elastic member; [0049] **151**—winding station; **152**—material-unloading station; [0050] **160**—cutter assembly; **161**—cutter driving member; **162**—cutter; **163**—heating rod; [0051] **170**—finishing assembly; **171**—finish compressing member; **1711**—bracket; **1712**—pressure driving member; **1713**—pressing wheel; **172**—finishing adsorption member; **1721**—adsorption driving member; **1722**—finishing adsorption plate.

DETAILED DESCRIPTION

[0052] The following description will clearly and completely describe the technical solutions in the embodiments of the present disclosure with reference to the accompanying drawings of the embodiments of the present disclosure. Obviously, the described embodiments are only part of the embodiments of the present disclosure, not all of them. Based on the embodiments in the present disclosure, all other embodiments obtained by those skilled in the art without creative labor fall within the scope of protection of the present disclosure.

[0053] In the present disclosure, an orientation or positional relationship indicated by terms “up”, “down”, “left”, “right”, “front”, “back”, “top”, “bottom”, “inner”, “outer”, “vertical”, “horizontal”, “transverse”, “longitudinal”, etc., is based on the orientation or positional relationship shown in the accompanying drawings. These terms are mainly intended to better describe the present disclosure and its embodiments, rather than to limit that the indicated devices, components, or parts must have specific orientations or be constructed and operated in specific orientations.

[0054] Moreover, some of these terms may also be used to indicate other meanings besides indicating orientation or positional relationships. For example, the term “up” may also indicate a certain attachment relationship or connection relationship in certain cases. Those skilled in the art may understand the specific meanings of these terms in the present disclosure according to specific situations.

[0055] In addition, terms “installed”, “disposed”, “provided with”, “connected”, and “communicated” should be interpreted in a broad sense. For example, it can be a fixed connection, a detachable connection, or an integrated construction; it can be a mechanical connection or an electrical connection; it can be directly connected, or indirectly connected through an intermediate medium, and it can be an internal communication between two devices, elements or parts. For those of ordinary skill in the art, specific meanings of the above terms in the present disclosure can be understood according to specific situations.

[0056] In addition, terms such as “first”, “second”, etc., are mainly used to distinguish different devices, elements, or parts (which may be the same or different in specific types and structures), and do not imply or suggest relative importance and quantity of the indicated devices, elements, or parts. Unless otherwise stated, the term “plurality” means two or more.

[0057] As described in the background of the present disclosure, in the relevant technology, a

clamping member, such as an inner clamping needle, is provided on a winding needle. This inner clamping needle can clamp and fix the initial winding end of the material strip onto the winding needle before the material strip is wound onto the winding needle, thereby preventing the material strip from sliding relative to the winding needle. In the case that the material strip can be prevented from sliding relative to the winding needle by the inner clamping needle, the material strip can be wound neatly onto the winding needle when the winding needle base starts to drive the winding needle to rotate so as to wind the material strip onto the winding needle.

[0058] However, when winding a small-sized cell, the winding needle that is adapted to the small cell must also be small. Due to size constraints, it is not feasible to provide a clamping member on the winding needle to fix the initial winding end of the material strip. Consequently, when the winding needle base starts to drive the winding needle to rotate, the material strip may slip relative to the winding needle. This causes the subsequent winding to be disordered and uneven, resulting in suboptimal winding performance.

[0059] In order to solve the technical problems mentioned in the background, the present disclosure provides a winding device provided with a restraining member provided outside the winding needle gap. The restraining member can fix the material strip protruding from the winding needle gap with a preset force, and the preset force is less than the winding tension applied to the material strip when the winding needle base drives the winding needle to rotate. When the winding needle base drives the winding needle to rotate relative to the mounting table, it is possible to prevent the material strip protruding from the winding needle gap from sliding relative to the winding needle, and at the same time to ensure that the material strip protruding from the winding needle gap is wound onto the winding needle, thereby improving the winding performance of the material strip. Additionally, since the restraining member is provided outside the winding needle gap, the winding device can be adapted for winding a miniaturized cell.

[0060] The present disclosure is described in detail below by way of specific embodiments:

[0061] Referring to FIGS. 1 and 2, the present disclosure provides a winding device **100**. The winding device **100** includes a material-placing mechanism **110**, a mounting table **120**, a winding member **130**, and a restraining member **140**, and the material-placing mechanism **110** is configured to convey a material strip **10**; the winding member **130** includes a winding needle base **131** and a plurality of winding needles **132** provided on the winding needle base **131**, the winding needle base **131** is rotatably provided on the mounting table **120**, with a winding needle gap **133** formed between the plurality of winding needles **132** and configured to allow the material strip **10** to pass through; the restraining member **140** is provided outside the winding needle gap **133** and configured to fix the material strip **10** protruding from the winding needle gap **133** with a preset force F_1 , a winding tension applied to the material strip **10** when the winding needle base **131** drives the winding needle **132** to rotate is F_2 , and F_1 is less than F_2 so as to wind both a preset length of the material strip **10** and the material strip **10** protruding from the winding needle gap **133** onto the winding needle **132** when the winding needle base **131** drives the winding needle **132** to rotate.

[0062] In the present embodiment, since the restraining member **140** can fix the material strip **10** protruding from the winding needle gap **133** with a preset force, and the preset force is less than the winding tension applied to the material strip **10** when the winding needle base **131** drives the winding needle **132** to rotate, when the winding needle base **131** drives the winding needle **132** to rotate relative to the mounting table **120**, it is possible to prevent the material strip **10** protruding from the winding needle gap **133** from sliding relative to the winding needle **132**, and at the same time to ensure that the material strip **10** protruding from the winding needle gap **133** is wound onto the winding needle **132**, thereby ensuring the neatness of the material strip **10** wound onto the winding needle **132** and improving the winding performance of the material strip **10**. Additionally, since the restraining member **140** is provided outside the winding needle gap **133**, the winding device **100** can be adapted for winding a small-sized cell.

[0063] In practical applications, the restraining member **140** first fixes the material strip **10** protruding from the winding needle gap **133** with a preset force, and then the winding needle base **131** drives the winding needle **132** to rotate relative to the mounting table **120**, thereby winding both the preset length of the material strip **10** conveyed by the material-placing mechanism **110** and the material strip **10** protruding from the winding needle gap **133** onto the winding needle **132** to complete the initial winding of the cell.

[0064] In some possible embodiments, as shown in FIGS. **1** and **3**, the restraining member **140** can move towards or away from the winding needle **132** so as to fix the material strip **10** protruding from the winding needle gap **133** with a preset force or release it.

[0065] Specifically, when the restraining member **140** is moved in a direction close to the winding needle **132**, it can fix the material strip **10** protruding from the winding needle gap **133** with the preset force. When the restraining member **140** is moved in a direction away from the winding needle **132**, it releases the material strip **10** to avoid interference with the winding of the winding needle **132**.

[0066] Thus, by allowing the restraining member **140** to move towards or away from the winding needle **132** to fix the material strip **10** protruding from the winding needle gap **133** with a preset force, it is possible to dispose the restraining member **140** flexibly, and to prevent the restraining member **140** from interference with the winding of the material strip **10** in the winding process of the winding needle **132**.

[0067] The structure of the restraining member **140** can take various forms. Herein, in some possible structures, referring to FIGS. **1** and **3**, the restraining member **140** includes a first driving member **1411** and an adsorption member **141** connected to the first driving member **1411**. The first driving member **1411** is configured to drive the adsorption member **141** to move towards or away from the winding needle **132**, and the adsorption member **141** is configured to adsorb and fix the material strip **10** protruding from the winding needle gap **133**.

[0068] Thus, by adsorbing and fixing the material strip **10** protruding from the winding needle gap **133** through the adsorption member **141**, both the material strip **10** conveyed by the material-placing mechanism **110** and the material strip **10** protruding from the winding needle gap **133** can be wound onto the winding needle **132** when the winding needle base **131** drives the winding needle **132** to rotate relative to the mounting table **120**. Additionally, by using the adsorption member **141** to fix the material strip **10** protruding from the winding needle gap **133**, compared to a winding solution that uses a clamping member fixes the material strip **10** wound onto the winding needle **132**, it is possible to avoid damage to the surface of the material strip **10** and further ensure the winding performance of the cell.

[0069] To enable the adsorption plate to adsorb and fix the material strip **10**, in some possible embodiments, referring to FIG. **4**, the adsorption member **141** is an adsorption plate, which is provided with a plurality of first adsorption holes **1412** on a side of the adsorption plate facing the material strip **10**.

[0070] Thus, by providing the plurality of first adsorption holes **1412** on the side of the adsorption plate facing the material strip **10**, it is possible to ensure the effect of adsorbing and fixing the material strip **10** by the adsorption plate.

[0071] In another possible structure of the restraining member **140**, referring to FIGS. **5**, **6** and **7**, the restraining member **140** includes a second driving member **1421** and a pressure roller **142** rotatably connected to the second driving member **1421**. The second driving member **1421** is configured to drive the pressure roller **142** to move towards or away from the winding needle **132**, and the pressure roller **142** is configured to press the material strip **10** protruding from the winding needle gap **133** onto the winding needle **132** to fix the material strip **10** by pressing.

[0072] Thus, by pressing the material strip **10** protruding from the winding needle gap **133** onto the winding needle **132**, it is possible to prevent the material strip **10** from sliding relative to the winding needle **132** during the winding needle base **131** driving the winding needle **132** to rotate

relative to the mounting table **120**, such that both the material strip **10** conveyed by the material-placing mechanism **110** and the material strip **10** protruding from the winding needle gap **133** are wound onto the winding needle **132** during the winding needle base **131** driving the winding needle **132** to rotate relative to the mounting table **120**. Additionally, by fixing the material strip **10** protruding from the winding needle gap **133** by pressing, in a way that uses the pressure roller **142** fixes the material strip **10** onto the winding needle **132** by pressing, compared to a winding solution that uses a clamping member fixes the material strip **10** wound onto the winding needle **132**, it is possible to avoid damage to the surface of the material strip **10** and further ensure the winding performance of the cell.

[0073] It should also be noted that both the first driving member **1411** and the second driving member **1421** can be linear motors or pneumatic cylinders.

[0074] Since the pressure roller **142** is rigid, to avoid excessive pressing of the material strip **10** by the pressure roller **142**, in some possible embodiments, referring to FIG. 7, the restraining member **140** also includes a mounting base **1422** connected to the second driving member **1421** and an elastic member **1423** provided on the mounting base **1422**, with the pressure roller **142** provided on the elastic member **1423**.

[0075] Since the pressure roller **142** is provided on the mounting base **1422** via the elastic member **1423**, when the pressure roller **142** presses the material strip **10** onto the winding needle **132**, the elastic member **1423** can buffer the pressing force between a part of the pressure roller **142** and the winding needle **132**, thereby preventing excessive pressing of the material strip **10** by the pressure roller **142**.

[0076] Herein, the elastic member **1423** can be a spring, a rubber block, or similar structures.

[0077] In some possible embodiments, referring to FIG. 8, the mounting table **120** includes a rotating plate **121**, which is rotatable about its central axis; the winding member **130** includes a first winding member **134** and a second winding member **135**. The winding device **100** has a winding station **151** and a material-unloading station **152**. When the first winding member **134** winds a preset length of the material strip **10** at the winding station **151**, the rotating plate **121** is configured to drive the first winding member **134** to rotate to the material-unloading station **152**, and to drive the second winding member **135** to rotate to the winding station **151**. The winding needle base **131** of the second winding member **135** is also configured to drive the winding needle **132** to slide and protrude out relative to the rotating plate **121** along the axial direction of the winding needle **132**, so as to enable the material strip **10** to be located in the winding needle gap **133** of the second winding member **135**. The winding device **100** also includes a cutter assembly **160**, which is configured to cut off the material strip **10** between the first winding member **134** and the second winding member **135**, and the restraining member **140** is configured to fix the material strip **10** which protrudes from the winding needle gap **133** and has been cut off.

[0078] In practical applications, the first winding member **134** is first made to be positioned at the winding station **151**, and then the restraining member **140** is made to fix a section of the material strip **10** protruding from the winding needle gap **133** of the first winding member **134**, so as to prevent the section of the material strip **10** from slipping relative to the winding needle **132** during rotation of the winding needle **132** of the first winding member **134**, and at the same time, to allow the section of the material strip **10** protruding from the winding needle gap **133** of the first winding member **134** to be wound onto the winding needle **132** of the first winding member **134**. When the first winding member **134** winds the preset length of the material strip **10** conveyed by the material-placing mechanism **110** onto the first winding member **134** at the winding station **151**, the rotating plate **121** is rotated, so as to drive the first winding member **134** to rotate to the material-unloading station **152**, and to drive the second winding member **135** to rotate to the winding station **151**. When the second winding member **135** is at the winding station **151**, the rotation of the rotating plate **121** is stopped, and meanwhile, the winding needle base **131** of the second winding member **135** is made to drive the winding needle **132** to slide and protrude out relative to the

rotating plate **121** along the axial direction of the winding needle **132**, so as to locate the material strip **10** in the winding needle gap **133** of the second winding member **135**. Then, the restraining member **140** is made to fix the material strip **10** protruding from the winding needle gap **133** of the second winding member **135** with a preset force; subsequently, the cutter assembly **160** is adopted to cut off the material strip **10** between the first winding member **134** and the second winding member **135**; finally, the winding needle base **131** of the second winding member **135** is made to drive its corresponding winding needle **132** to rotate relative to the rotating plate **121**, so as to wind both the preset length of the material strip **10** conveyed by the material-placing mechanism **110** and the material strip **10** protruding from the winding needle gap **133** of the second winding member **135** onto the winding needle **132** of the second winding member **135**.

[0079] Additionally, similarly, after the winding needle **132** of the second winding member **135** completes the winding and the first winding member **134** finishes unloading, the rotating plate **121** is rotated again to rotate the first winding member **134** to the winding station **151** and to rotate the second winding member **135** to the material-unloading station **152** for a further winding of the first winding member **134**, and so on, therefore performing a plurality of windings of the first winding member **134** and the second winding member **135**.

[0080] It can be seen that when the winding member **130** includes both the first winding member **134** and the second winding member **135**, and the winding of the cell can be completed on the first winding member **134**, by rotating the first winding member **134** to the material-unloading station **152** and rotating the second winding member **135** to the winding station **151** via the rotating plate **121**, it is possible to reduce the winding time for the first winding member **134** and the second winding member **135** and improve the winding efficiency of the cell.

[0081] In some possible embodiments, referring to FIGS. **8** and **9**, the cutter assembly **160** includes a cutter driving member **161** and a cutter **162** connected to the cutter driving member **161**. The cutter driving member **161** is configured to drive the cutter **162** to move towards the material strip **10**, so as to cut off the material strip **10** between the first winding member **134** and the second winding member **135**.

[0082] Thus, by using the cutter driving member **161** to drive the cutter **162** to cut off the material strip **10** between the first winding member **134** and the second winding member **135**, compared to manually cutting off the material strip **10** between the first winding member **134** and the second winding member **135**, the winding efficiency of the winding device **100** is further improved.

[0083] Herein, the cutter driving member **161** can be a pneumatic cylinder or a linear motor, among others.

[0084] Alternatively, the material strip **10** can be two layers of diaphragms that are stacked together, or alternatively, the material strip **10** can be two layers of diaphragms with a pole piece sandwiched between them.

[0085] During the winding of the cell, when the material strip **10** includes two layers of diaphragms, and the cutter **162** cuts off the material strip **10** between the first winding member **134** and the second winding member **135**, the two layers of diaphragms are highly prone to displacement, which can affect the winding performance of the wound cell. Based on this, in some possible embodiments, referring to FIG. **9**, the cutter **162** is provided with a heating rod **163** therein, which is configured to heat the cutter **162**.

[0086] Since the heating rod **163** can heat the cutter **162**, when the cutter **162** cuts off the material strip **10**, it is possible to bond the two layers of diaphragms together, thereby preventing the displacement of the diaphragms on both sides and thus ensuring the winding performance of the wound cell.

[0087] When the cutter **162** cuts off the material strip **10**, since the material strip **10** is flexible, to ensure the cutting effect of the cutter **162**, in some possible embodiments, referring to FIG. **10**, the mounting table **120** also includes a connection shaft **123** and an abutting member **122** provided on the connection shaft **123**, the abutting member **122** is configured to provide an abutting force to the

material strip **10** when the cutter **162** cuts off the material strip **10** between the first winding member **134** and the second winding member **135**, and the rotating plate is fixed to the connection shaft **123**.

[0088] Since the abutting member **122** is provided on the connection shaft **123** and provides an abutting force to the material strip **10** when the cutter **162** cuts off the material strip **10**, the abutting force of the cutter **162** applied to the material strip **10** and the abutting force of the abutting member **122** applied to the material strip **10** are in opposite directions, so as to prevent the material strip **10** from moving in the direction of the abutting force of the cutter **162** during the cutting process of the cutter **162**, thereby ensuring the cutting effect of the cutter **162** to the material strip **10**.

[0089] Furthermore, in some possible embodiments, referring to FIG. **10**, the abutting member **122** is provided with an avoidance groove **1221**, which is configured to avoid the cutter **162**.

[0090] Therefore, by providing the avoidance groove **1221**, it is possible to avoid the cutter **162** when cutting the material strip **10**, so that the cutter **162** can penetrate the material strip **10** and be inserted into the avoidance groove **1221**, thereby ensuring the cutting effect of the cutter **162**.

[0091] In some possible embodiments, referring to FIG. **11**, the abutting member **122**, on a side facing the material strip **10**, is provided with a first adsorption plate **1222** and a second adsorption plate **1223**, there is an avoidance gap between the first adsorption plate **1222** and the second adsorption plate **1223**, the avoidance gap forms the avoidance groove **1221**, and the first adsorption plate **1222** and the second adsorption plate **1223** are configured to adsorb the material strip **10**.

[0092] Specifically, both the first adsorption plate **1222** and the second adsorption plate **1223** are provided with a plurality of second adsorption holes **1224**.

[0093] Thus, when the cutter **162** cuts the material strip **10**, the first adsorption plate **1222** and the second adsorption plate **1223** are first made to adsorb and fix the material strip **10** onto the abutting member **122**, and then, the cutter driving member **161** drives the cutter **162** to move towards the abutting member **122** until the cutter **162** penetrates the material strip **10** and is inserted into the avoidance groove **1221**, thereby completing the cutting of the material strip **10** by the cutter **162**. It can be seen that by providing the first adsorption plate **1222** and the second adsorption plate **1223**, it is possible to ensure the stability of providing the material strip **10** when the cutter **162** cuts the material strip **10**, thereby ensuring the cutting effect of the cutter **162** to the material strip **10**.

[0094] In some possible embodiments, referring to FIGS. **12** and **13**, the winding device **100** further includes a finishing assembly **170**, the finishing assembly **170** includes a finish compressing member **171** provided adjacent to the material-unloading station **152**, and the finish compressing member **171** is configured to press the material strip **10**, which is adjacent to the material-unloading station **152** and has been cut off, onto the winding member **130** at the material-unloading station **152**.

[0095] Herein, the above finish compressing member **171** is configured to press the material strip **10**, which is adjacent to the material-unloading station **152** and has been cut off, onto the winding member **130** at the material-unloading station **152**. It should be understood that when the first winding member **134** is at the material-unloading station **152**, the finish compressing member **171** is configured to press a section of the material strip **10** wound onto the winding needle **132** of the first winding member **134** to the winding needle **132** of the first winding member **134**; alternatively, when the second winding member **135** is at the material-unloading station **152**, the finish compressing member **171** is configured to press a section of the material strip **10** wound onto the winding needle **132** of the second winding member **135** to the winding needle **132** of the second winding member **135**.

[0096] Thus, when the cutter **162** cuts off the material strip **10** between the first winding member **134** and the second winding member **135**, by providing the finish compressing member **171**, it is possible to ensure the winding performance of a section of the material strip **10** wound onto the first winding member **134** or the second winding member **135**.

[0097] Specifically, the finish compressing member **171** includes a bracket **1711**, a pressure driving member **1712** provided on the bracket **1711**, and a pressing wheel **1713** connected to the pressure driving member **1712**. When the pressure driving member **1712** drives the pressing wheel **1713** to press onto the winding needle **132** at the material-unloading station **152**, on one hand, the pressing wheel **1713** can press the material strip **10** to wind it onto the winding needle **132** at the material-unloading station **152**, thereby ensuring the winding performance of the material strip **10**; on the other hand, since the pressing wheel **1713** rolls against the material strip **10**, it is possible to reduce the friction between the material strip **10** and the pressing wheel **1713**, further ensuring the winding performance of the material strip **10**.

[0098] In some possible embodiments, referring to FIGS. **12** and **13**, the finishing assembly **170** further includes a finishing adsorption member **172** provided adjacent to the material-unloading station **152**, and the finishing adsorption member **172** is configured to adsorb the material strip **10**, which is adjacent to the material-unloading station **152** and has been cut off, so as to prevent a free end of the material strip **10** from swinging.

[0099] By providing the finishing adsorption member **172**, it is possible to adsorb a section of the material strip **10** wound onto the winding needle **132** at the material-unloading station **152**, thereby improving the winding performance of the first winding member **134**.

[0100] Specifically, the finishing adsorption member **172** includes an adsorption driving member **1721** provided on the bracket **1711** and a finishing adsorption plate **1722** connected to the adsorption driving member **1721**. When the cutter **162** cuts off the material strip **10** between the first winding member **134** and the second winding member **135**, the adsorption driving member **1721** drives the finishing adsorption plate **1722** to move towards the material strip **10**, so as to adsorb a section of the material strip **10** wound onto the winding needle **132** of the first winding member **134**, thereby preventing the material strip **10** that has been cut off from swinging and thus ensuring the winding performance of the cell.

[0101] Finally, it should be noted that the above embodiments are only illustrative of the technical solutions of the present disclosure and are not intended to limit the scope thereof. Although the present disclosure has been described in detail with reference to the foregoing embodiments, a person of ordinary skills in the art will appreciate that various modifications may be made to the technical solutions disclosed in the foregoing embodiments, or that certain technical features may be substituted on an equivalent basis, either partially or entirely. Such modifications or substitutions do not depart from the essential spirit and the scope of the technical solutions of the embodiments of the present disclosure.

Claims

1. A winding device, comprising: a material-placing mechanism configured to convey a material strip; a mounting table; a winding member comprising a winding needle base and a plurality of winding needles provided on the winding needle base, the winding needle base being rotatably provided on the mounting table, wherein a winding needle gap is formed between the plurality of winding needles, such that the material strip passes therethrough; and a restraining member provided outside the winding needle gap and configured to fix the material strip protruding from the winding needle gap with a preset force F_1 , wherein the winding needle base is configured to drive the winding needle to rotate with a winding tension F_2 applied to the material strip, and where F_1 is less than F_2 , such that both a first portion of the material strip having a preset length and a second portion of the material strip protruding from the winding needle gap are wound onto the winding needle.

2. The winding device according to claim 1, wherein the restraining member is movable towards or away from the winding needle to fix the material strip protruding from the winding needle gap with the preset force or release it.

3. The winding device according to claim 1, wherein the restraining member comprises a first driving member and an adsorption member connected to the first driving member, the first driving member is configured to drive the adsorption member to move towards or away from the winding needle, and the adsorption member is configured to adsorb and fix the material strip protruding from the winding needle gap.
4. The winding device according to claim 3, wherein the adsorption member comprises an adsorption plate, which is provided with a plurality of first adsorption holes on a side of the adsorption plate facing the material strip.
5. The winding device according to claim 1, wherein the restraining member comprises a second driving member and a pressure roller rotatably connected to the second driving member, the second driving member is configured to drive the pressure roller to move towards or away from the winding needle, the material strip protruding from the winding needle gap is pressed onto the winding needle to fix the material strip by pressing.
6. The winding device according to claim 5, wherein the restraining member further comprises a mounting base connected to the second driving member, and an elastic member provided on the mounting base, with the pressure roller provided on the elastic member.
7. The winding device according to claim 1, wherein the mounting table comprises a rotating plate, which is rotatable along a central axis thereof; the winding member comprises a first winding member and a second winding member; the winding device further comprises a cutter assembly configured to cut off the material strip between the first winding member and the second winding member, and the restraining member is configured to fix the material strip which protrudes from the winding needle gap and has been cut off.
8. The winding device according to claim 7, wherein the cutter assembly comprises a cutter driving member and a cutter connected to the cutter driving member, and the cutter driving member is configured to drive the cutter to move towards the material strip to cut off the material strip between the first winding member and the second winding member.
9. The winding device according to claim 8, wherein the cutter is provided with a heating rod therein configured to heat the cutter.
10. The winding device according to claim 9, wherein the material strip comprises two layers of diaphragms, which are bonded together when the cutter (162) cuts the material strip (10) off.
11. The winding device according to claim 8, wherein the mounting table further comprises a connection shaft and an abutting member provided on the connection shaft, the abutting member is configured to provide an abutting force to the material strip when the cutter cuts off the material strip between the first winding member and the second winding member, and the rotating plate is fixed to the connection shaft.
12. The winding device according to claim 11, wherein the abutting member is provided with an avoidance groove configured to avoid the cutter.
13. The winding device according to claim 11, wherein the abutting member, on a side facing the material strip, is provided with a first adsorption plate and a second adsorption plate, there is an avoidance gap between the first adsorption plate and the second adsorption plate to form the avoidance groove, and the first adsorption plate and the second adsorption plate are configured to adsorb the material strip.
14. The winding device according to claim 1, further comprises a finishing assembly, and a material-unloading station; wherein the finishing assembly comprises a finish compressing member provided adjacent to the material-unloading station, the finish compressing member is configured to press the material strip onto the winding member at the material-unloading station, and the material strip is adjacent to the material-unloading station and has been cut off.
15. The winding device according to claim 14, wherein the finishing assembly further comprises a finishing adsorption member, the finishing adsorption member is configured to adsorb the material

strip to prevent a free end of the material strip from swinging, and the material strip is adjacent to the material-unloading station and has been cut off.
